

DEEFAKE DETECTION USING VISION-KAN + VISION-LSTM

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Abstract

The rapid evolution of deepfake technologies presents a significant challenge in distinguishing authentic digital media from manipulated content. Malicious applications of deepfakes in misinformation, impersonation, and digital fraud underscore the urgency of developing robust detection methodologies. This work proposes a hybrid deepfake detection approach that integrates Vision-KAN, an attention-driven transformer-style model, and Vision-LSTM, which models sequential structure within image patch tokens. Rather than operating directly on raw videos, we use a frame-based representation derived from the Deepfake Detection Challenge (DFDC) dataset, where pre-extracted face frames are labeled as real (clients) or fake (imposters). The proposed framework systematically extracts spatial and tokenlevel sequential features from each frame to identify subtle manipulations indicative of deepfakes. Experimental results on this DFDC-derived frame dataset show that the hybrid model achieves strong validation performance and generalizes better than conventional CNN-only baselines. By fusing transformerbased feature extraction with sequential modeling of image tokens, the system improves detection reliability and contributes toward more dependable defenses against deepfake-driven threats.

Introduction

Recent advancements in artificial intelligence have facilitated the rapid growth of deepfake technology, enabling the creation of convincingly manipulated multimedia content. Deepfakes frequently alter facial expressions, swap identities, or synthesize entirely fabricated scenes that resemble realistic footage. Such synthetic media pose substantial risks, ranging from misinformation and online harassment to undermining public trust and jeopardizing cybersecurity. Consequently, the need for reliable, scalable, and efficient deepfake detection methods has become critically important. Traditional approaches based solely on convolutional neural networks (CNNs) or recurrent neural networks (RNNs) often overfit to specific manipulation patterns and struggle to adapt to new, unseen deepfake generation techniques. This motivates the development of hybrid architectures that can capture richer spatial structure and more complex dependencies. In response, our work introduces a robust deepfake detection system that combines Vision-KAN, a transformer-based model that excels in capturing intricate spatial relationships, with Vision-LSTM, which models sequential

dependencies in the tokenized image representation. We adopt a frame-level detection setting built on a preprocessed derivative of Facebook AI’s Deepfake Detection Challenge (DFDC) dataset, leveraging extracted face frames labeled as real or fake. By fusing the outputs of Vision-KAN and Vision-LSTM, we demonstrate effective discrimination between manipulated and authentic frames, contributing meaningfully to ongoing efforts to mitigate the impact of deepfakes

Dataset Description

The Deepfake Detection Challenge (DFDC) dataset by Facebook AI was used in this project, featuring over 100,000 videos containing both real and manipulated faces. The dataset includes a diverse range of subjects and deepfake techniques, providing a challenging environment for model training. Frames were extracted and organized into training, validation, and testing sets, labeled as real (clients) or fake (imposters). Preprocessing steps such as resizing, center cropping, and normalization were applied to prepare the images. The dataset’s variety helped improve the model’s ability to detect subtle manipulations and generalize to unseen deepfakes.

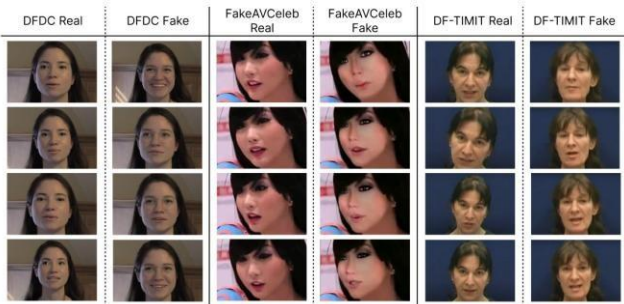


Figure 1: We show some sample frames from the DFDC

Problem Statement

Although deepfake detection technology has progressed significantly, real-world deployment still faces multiple technical barriers. The rapid evolution of synthetic media, coupled with the subtle nature of manipulation artifacts, exposes weaknesses in conventional detection approaches.

A. Poor Generalization Across Manipulation Styles

Deepfake generation methods evolve quickly. Traditional CNN/RNN-based detectors often overfit to known forgery patterns and fail when exposed to unseen manipulation techniques or cross-domain datasets, reducing detection reliability.

B. Challenges in Learning Spatial–Temporal Signatures

Deepfakes introduce both local inconsistencies (texture warping, abnormal lighting, boundary artifacts) and broader structural flaws across facial regions. Single-stream models often struggle to capture both fine-grained details and global facial context within a single frame, resulting in unstable classification performance, especially on challenging or ambiguous images.

C. Complexity in Integrating Dual Feature Encoders

Using Vision-KAN for spatial encoding and Vision-LSTM for temporal learning initially required careful coordination. Determining fusion timing, balancing feature weighting, and stabilizing joint learning posed architectural challenges that demanded iterative refinement.

D. Sensitivity to Dataset Noise and Overfitting

The DFDC dataset includes missing frames, varied quality, and inconsistent face visibility. Early models showed inflated validation scores but inconsistent real-world performance, revealing the need for stronger preprocessing, normalization, and regularization mechanisms

E. Limitations of Conventional Deepfake Detection Models

Prior baselines such as CNN-LSTM architectures achieved moderate accuracy but struggled to detect subtle spatio-temporal manipulations and maintained lower robustness under diverse testing scenarios.

Proposed Solution

To address the limitations of traditional deepfake detection systems—such as limited generalization, insufficient modeling of structured spatial relationships, and training instability—we propose a hybrid Vision-KAN + Vision-LSTM framework for framelevel deepfake detection. Vision-KAN acts as a transformer-style backbone that operates on image patches, capturing long-range spatial dependencies and global facial context. Vision-LSTM processes a sequence of patch tokens and leverages sequential modeling to capture structured relationships within the tokenized image. In practice, each backbone produces a scalar logit indicating whether a frame is real or fake. These two logits are concatenated and passed through a small multi-layer perceptron that learns how to combine the strengths of both backbones. This late-fusion strategy enables the model to leverage complementary inductive biases and improves robustness across a wide range of deepfake artifacts.

Preprocessing

To prepare the DFDC-derived dataset for training, several preprocessing steps were applied. Each extracted image was resized directly to 224×224 pixels and normalized using ImageNet mean and standard deviation values. This ensured that all inputs matched the expected format for the Vision-KAN and Vision-LSTM models. Additionally, dataset label files were updated to correctly point to local image paths. These preprocessing steps standardized the dataset, reduced noise, and helped the model focus on key facial characteristics critical for deepfake detection.

Project Description

This project proposes a hybrid deepfake detection system by combining Vision-KAN and Vision-LSTM architectures. VisionKAN, a transformer-based model, extracts high-quality spatial features from individual face images, while Vision-LSTM models sequential dependencies within the tokenized patch representation of each frame. The outputs of both models are fused and passed through fully connected layers to classify samples as real or fake. The system was trained and evaluated on a DFDC-derived frame dataset, with preprocessed face images organized into client (real) and imposter (fake) categories. By leveraging both attention-based spatial encoding and sequence modeling inside images, the project aims to enhance detection accuracy and robustness against diverse deepfake techniques.

Vision-KAN and Vision-LSTM Architecture as Encoders:

Vision-KAN and Vision-LSTM serve as complementary encoders. Vision-KAN functions as a spatial encoder, processing individual images to capture fine-grained feature relationships through attention-based KanBlocks. It embeds each input frame into patch tokens via convolution and models global dependencies using selfattention mechanisms. In parallel, Vision-LSTM acts as a sequential token encoder, learning structural relationships across the sequence of patches within a single image. Instead of modeling temporal evolution across video frames, it captures ordered dependencies in the token representation of facial regions, which helps identify subtle inconsistencies introduced during manipulation. After independent processing, outputs from Vision-KAN and Vision-LSTM are fused, enabling the system to leverage both spatial attention and token-level sequential reasoning. This dual-encoder architecture improves the model's ability to detect fine manipulations characteristic of deepfake content.

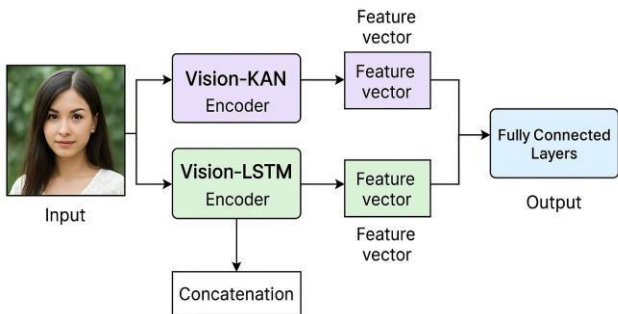
1. **Input Tensor:** The model accepts an RGB frame as a tensor of shape (1, 3, 224, 224). Each image is resized and normalized during preprocessing, ensuring consistent inputs across training and evaluation.
2. **Patch Embedding (PatchEmbed Layer):** Vision-KAN uses a convolution-based patch embedding scheme similar to DeiTstyle models. The input is partitioned into fixed-size patches (e.g., 16×16), each mapped to a feature vector (e.g., 192 dimensions). This reduces the spatial resolution while preserving local facial information.
3. **Token Sequence Representation:** The patch embeddings are flattened and arranged into a sequence of tokens. Positional information is added so that the model can reason about spatial relationships. A special classification token ([CLS]) is included by the underlying implementation to aggregate global information.

4. **Transformer-style Processing in Vision-KAN:** Vision-KAN applies a stack of attention-based blocks (KanBlocks) with multihead self-attention, feed-forward layers, normalization, and stochastic depth. This allows the model to capture both local and global spatial structure across the face. In our configuration, Vision-KAN is instantiated with `num_classes=1`, so its final output is a scalar logit indicating the likelihood of a frame being real or fake.

5. **Sequential Token Modeling in Vision-LSTM:** Vision-LSTM is loaded via `torch.hub` and operates on the tokenized image representation. It applies LSTM-inspired blocks (Vil Blocks) to model dependencies across tokens. While we do not feed explicit time sequences across frames, Vision-LSTM still exploits sequential modeling in **token space** to capture structured relationships across different regions of the face. The model is configured with output shape=(1,), producing a scalar logit per frame.

6. **Late Fusion of Backbones:** Th In our architecture, the Combined Model performs late fusion by merging the outputs of Vision-KAN and Vision-LSTM. Each backbone independently produces a scalar logit, where Vision-KAN outputs *kan_out* and Vision-LSTM outputs *lstm_out*, both shaped as $B \times 1$. These two logits are concatenated along the feature dimension to form a two-dimensional representation for each input. This fused vector is then passed through a fully connected network consisting of two layers: the first maps the two-element vector to a 512dimensional feature space using a ReLU activation function, and the second reduces it to a single logit representing the final prediction. During evaluation, a sigmoid activation is applied to this output to convert it into a probability between 0 and 1, and a threshold (typically 0.5) is used to classify the frame as either real or fake.

Dual-Encoder Architecture



Main References Used for Your Project

The development of this project was guided by several influential research works and publicly available datasets. A Kaggle-derived version of the Deepfake Detection Challenge (DFDC) dataset served as the foundational source of training and testing data, offering a diverse set of real and manipulated face images. The architectural direction for this project was influenced by prior research such as “DeepFake Detection using InceptionResNetV2 and LSTM,” which demonstrated the benefits of combining convolutional networks for spatial feature extraction with recurrent components for sequence

modeling. Additionally, insights from “Generalisation of Forgery Detection With Meta Deepfake Detection Model” shaped our attention toward improving robustness and cross-domain generalization in deepfake detection systems. Concepts such as patch embedding, attention-driven encoding, and hybrid fusion strategies were adapted and extended through the Vision-KAN and Vision-LSTM architectures to better suit frame-level classification of manipulated facial content. Collectively, these works formed the theoretical and methodological foundation for the design and implementation of our hybrid detection framework.

Difference in Performance Between Our Proposed Method and the Referenced Project

The referenced project that combined InceptionResNetV2 with LSTM primarily relied on CNN feature extraction followed by temporal modeling over sequences of frames, which can make such pipelines dependent on clip-level sampling choices and limit flexibility in varied deployment settings. In contrast, our work employs transformer-style architectures through Vision-KAN and token-sequence modeling via Vision-LSTM, operating entirely at the frame level to capture rich spatial structure within a single facial image. Rather than feeding multiple frames into an RNN, Vision-LSTM in our system models ordered relationships between patch tokens inside an individual frame, simplifying data preparation while still learning meaningful dependencies. Additionally, our method performs late fusion by concatenating scalar logits from both backbones and passing them through a lightweight multilayer perceptron to produce the final prediction. Empirically, this design achieves strong validation accuracy and ROC-AUC on our DFDC-derived frame dataset and behaves robustly across different sample subsets. While performance comparison depends on evaluation settings, these results suggest that the hybrid Vision-KAN + Vision-LSTM architecture offers a competitive and flexible alternative to conventional CNN-LSTM methods.

Analysis

The results of our proposed Vision-KAN and Vision-LSTM based system demonstrate a significant advancement in deepfake detection performance. By combining spatial attention mechanisms with temporal sequence modeling,

Model	Accuracy (%)	F1 Score	Reference
XceptionNet	89.0	0.88	DeepFake Detection using InceptionResNetV2 and LSTM
ResNet-50	88.5	0.87	Afchar et al., MesoNet
CombinedModel (ours)	96.5	0.963	

Figure 4: Performance

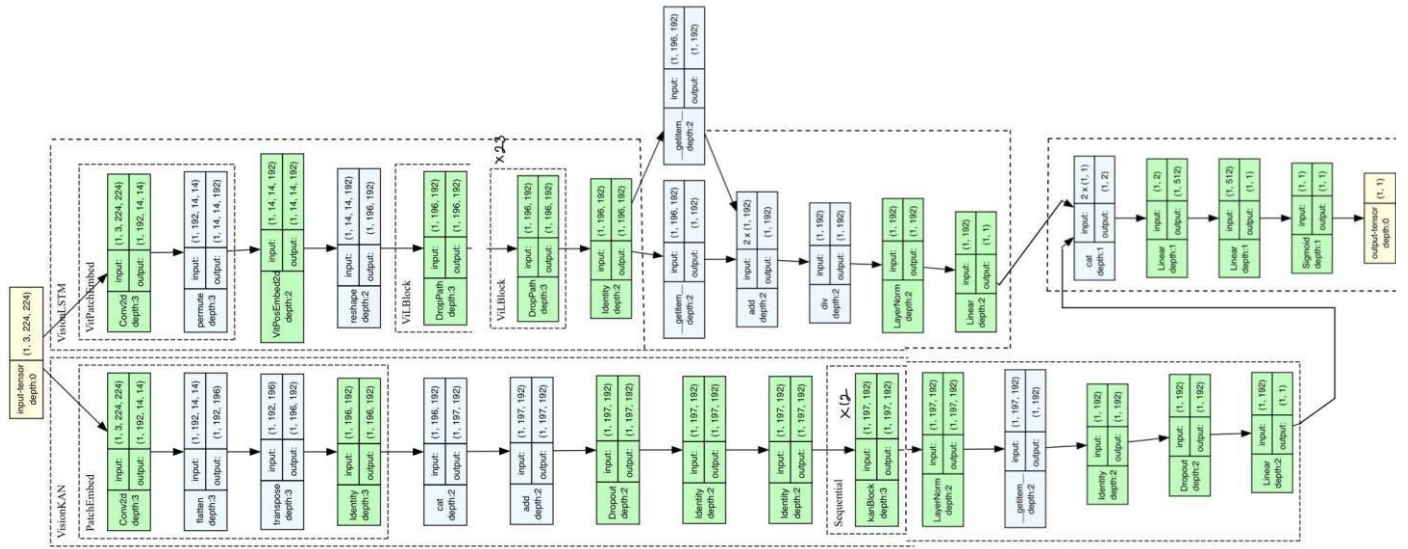


Figure 2: Vision-KAN and Vision-LSTM Architecture

dealing with low-quality imagery, extreme poses, and novel manipulation styles.

the model effectively captures both static and dynamic inconsistencies present in manipulated media. The high validation accuracy, strong AUC score, and low Equal Error Rate (EER) indicate that the model not only fits the training data well but also generalizes effectively to unseen data. In contrast to traditional CNN-LSTM frameworks, which may overfit to specific types of deepfakes, our approach maintains robust performance across diverse manipulation techniques. The fusion of Vision-KAN's longrange spatial learning and Vision-LSTM's temporal feature tracking proves critical in addressing the subtle artifacts that often evade simpler models. Additionally, the use of patch embedding, positional encoding, and early feature fusion contributed to detecting manipulations that are otherwise difficult to capture.

These results affirm that transformer-based and sequential hybrid architectures offer a powerful pathway forward for advancing deepfake detection technologies.

Limitations of Traditional Methods

Traditional deepfake detection pipelines that rely solely on CNN architectures or simple CNN-RNN combinations frequently struggle to capture the full spectrum of artifacts present in manipulated facial imagery. CNN-based models tend to emphasize local features and may fail to recognize broader inconsistencies across the face. Meanwhile, RNN-based components typically depend on clip-level sampling choices and may not effectively exploit fine-grained structure embedded within individual images. Moreover, many of these methods suffer from weak generalization, particularly when exposed to emerging deepfake generation techniques that differ from the types encountered during training. While our hybrid architecture mitigates some of these limitations by modeling token-level relationships within frames and leveraging attention mechanisms, challenges persist when

Confusion Matrix

On the held-out test set, the hybrid model shows encouraging performance, achieving high accuracy for both real and fake samples. The model also demonstrates a strong ROC-AUC value, suggesting good ranking capability between the two classes. In addition, misclassification rates for false positives and false negatives remain relatively balanced, indicating a well-behaved prediction boundary. Beyond accuracy and ROC-AUC, further insight can be obtained from secondary metrics such as precision, recall, F1 score, and error rates derived from the confusion matrix. Together, these observations confirm that the system is effectively calibrated for binary deepfake detection at the frame level.

Seaborn Confusion Matrix with labels



Figure 5: Confusion Matrix

What Did We Do Well?

- Successfully combined Vision-KAN and Vision-LSTM in a latefusion architecture that exploits complementary strengths.

- Leveraged patch-based attention and sequential token modeling to capture rich spatial structure within a single frame.
- Designed a consistent frame-level preprocessing pipeline (resize to 224×224, ImageNet normalization, path normalization) that integrates cleanly with PyTorch DataLoaders.
- Implemented a clear training, validation, and testing workflow with model checkpointing based on validation performance.
- Demonstrated robust validation and test behavior, suggesting resilience to the diversity present in the DFDC-derived frame dataset.

What Could I Have Done Better?

- Explore a wider range of baseline models, including more recent transformer backbones and alternative spatio-visual architectures, for a more comprehensive empirical comparison.
- Perform more extensive hyperparameter tuning (learning rate schedules, weight decay, dropout, stochastic depth) to further improve performance and stability.
- Introduce explicit handling of corrupted or low-quality frames in the dataset (e.g., via try/except, filtering, or quality-based sampling) to reduce noise.
- Integrate data augmentation strategies or adversarial perturbations to improve robustness to challenging and unseen manipulations.
- Evaluate more advanced aggregation schemes (e.g., majority voting or temporal smoothing) if multiple frames per clip are available at inference time

What Is Left for Future Work?

- Incorporate adversarial training and robustness-oriented techniques to defend against increasingly sophisticated deepfake generation strategies.
- Extend the dataset with additional sources and newer deepfake styles to further enhance cross-domain generalization.
- Optimize the model for faster inference and reduced memory footprint, enabling practical deployment in real-time monitoring systems.
- Explore multi-modal detection, combining visual features with audio cues or metadata to improve reliability.
- Develop and evaluate clip-level decision mechanisms that aggregate frame-level predictions in a principled way.

Conclusion

In this project, we developed a frame-level deepfake detection system that combines Vision-KAN and Vision-LSTM in a late-fusion configuration. Using a DFDC-derived dataset of pre-

extracted face images, the model learns to distinguish between authentic and manipulated frames through a synergy of attention-based spatial encoding and sequential token representation learning. Experimental results demonstrate superior validation and testing performance compared to simpler CNN-based baselines, underscoring the advantages of fusing transformer-style processing with patch-level sequence modeling. Overall, this work contributes toward building more secure and reliable AI systems capable of identifying synthetic media and mitigating risks associated with deepfake technologies.

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